DP Biology – C2.2 (HL extension)

Action potentials and synapses

- How do neurons produce electrical signals and pass them along?
- How do neurons affect other cells?

HL statements C2.2.8–16. Textbook pp. 479–489.

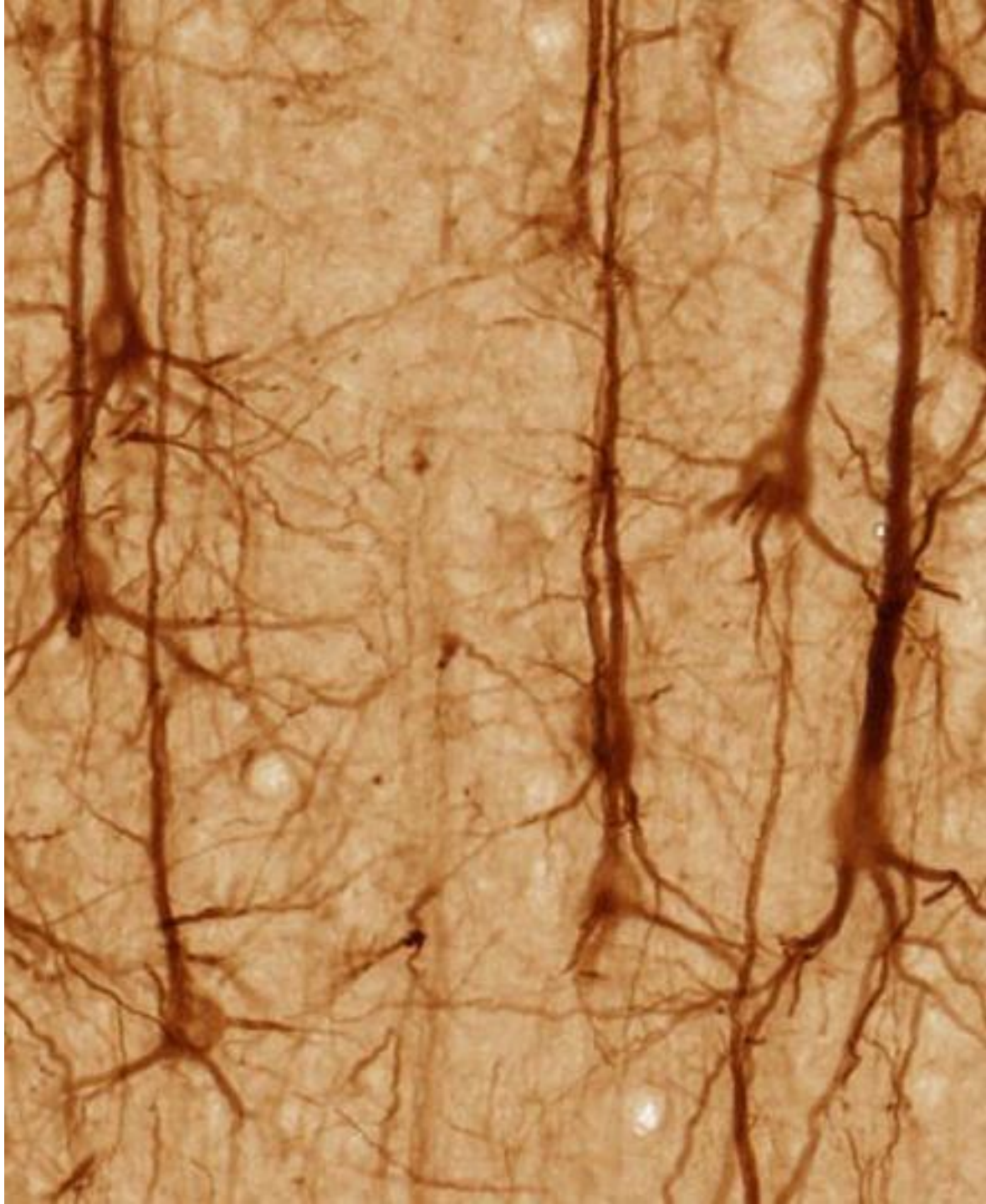
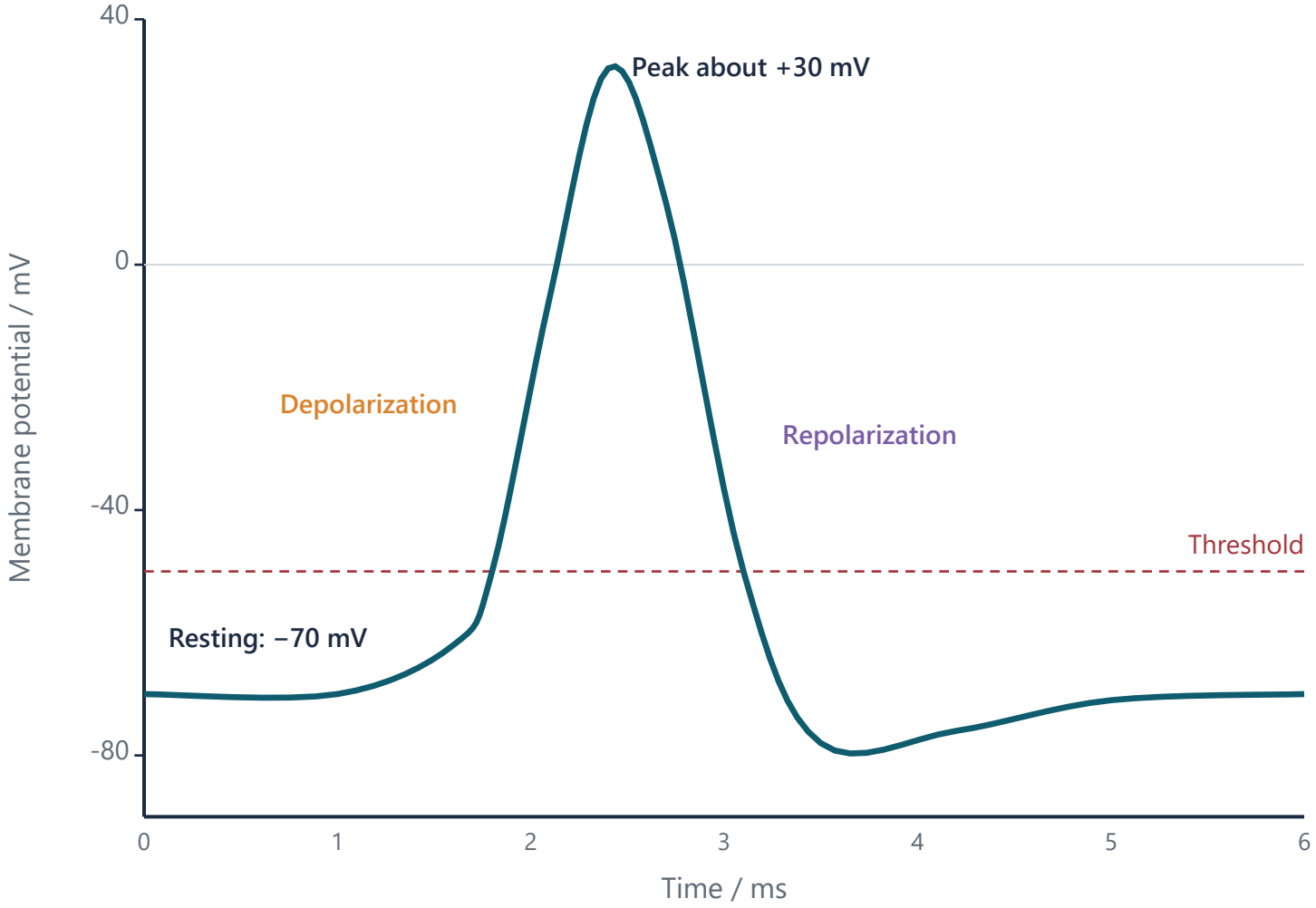


Image: neurons in the cerebral cortex, stained brown. UC Regents Davis campus, CC BY 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C2.2.8–10	Explain depolarization and repolarization, how local currents propagate an impulse, and read oscilloscope traces
2	C2.2.11–14	Explain saltatory conduction, the effects of drugs at synapses, inhibitory potentials and summation
3	C2.2.15–16	Explain how pain is detected, and consciousness as an emergent property

An action potential



An **action potential** is a quick reversal of the charge across the membrane, then a return to rest.

It lasts about **2 to 3 ms**.

It only starts if the membrane reaches the **threshold** (about -50 mV): all or nothing.

What the channels do

C2.2.8

Threshold reached → **Na^+ channels open: Na^+ rushes in** → Inside becomes positive: depolarization → Na^+ channels close; K^+ channels open → K^+ rushes out: repolarization

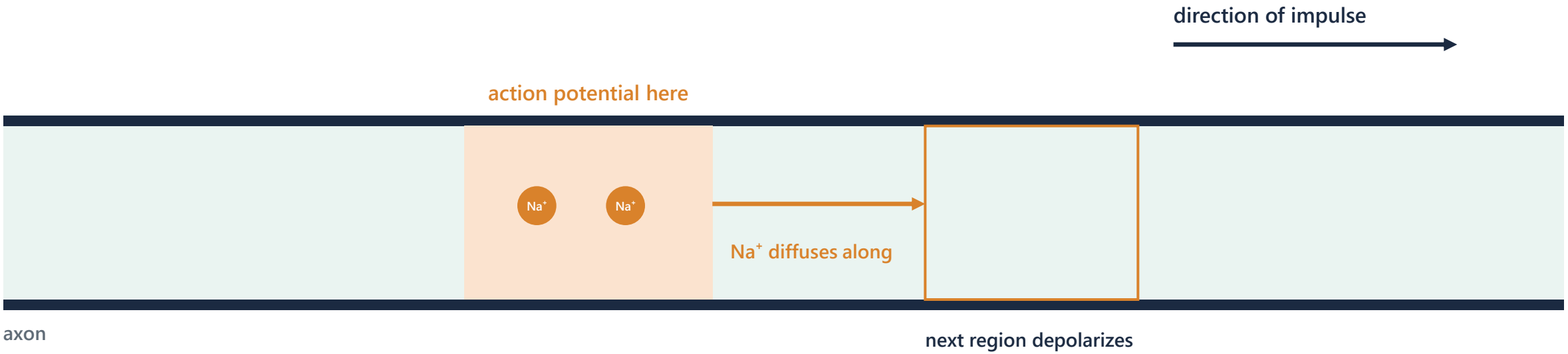


Voltage-gated

These channels open and close in response to the voltage across the membrane. Each ion simply diffuses down its gradient; no ATP is used during the action potential itself.

How the impulse moves along

C2.2.9

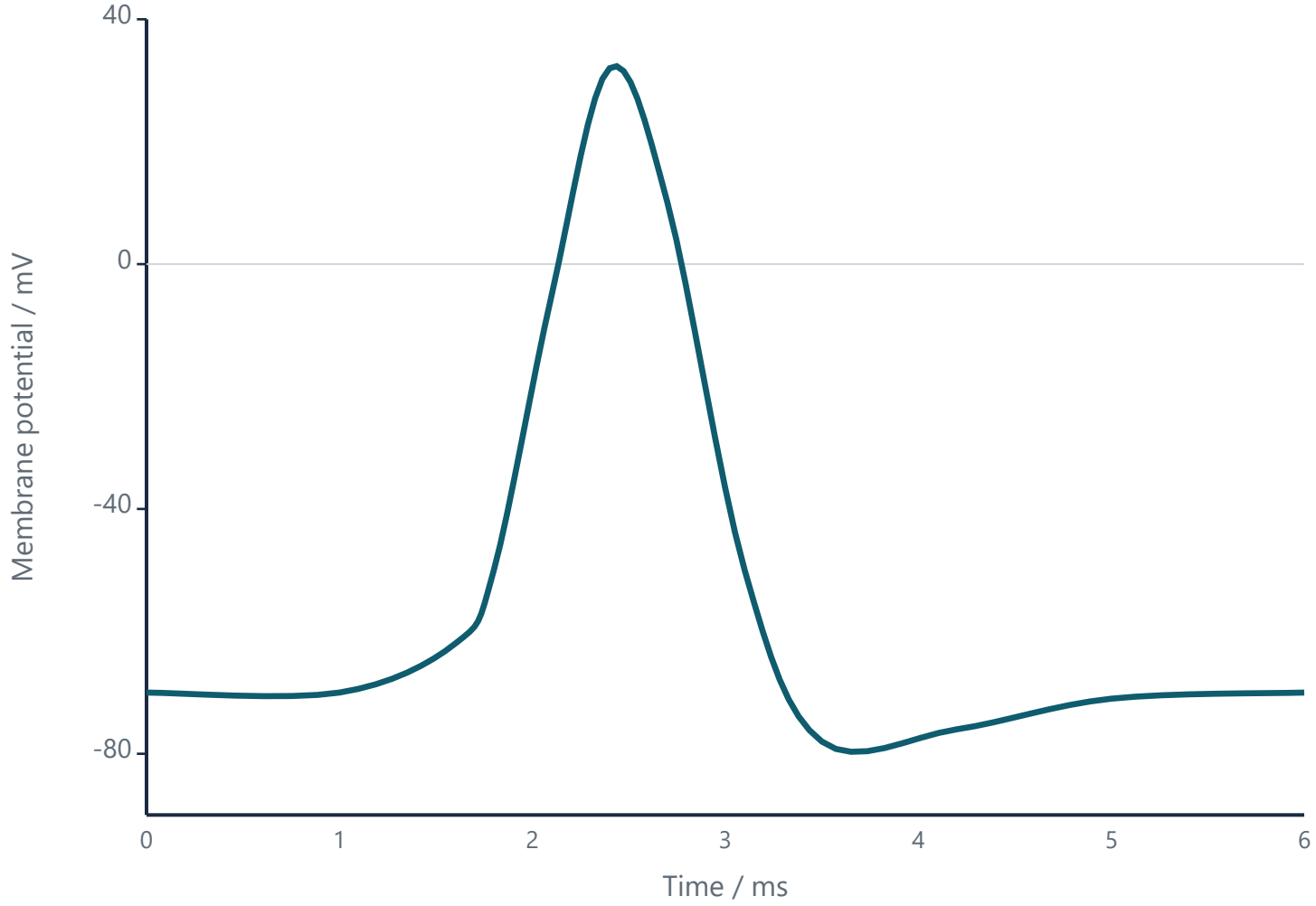


Na^+ that entered at the action potential diffuses along the axon: a **local current**. It raises the next region to threshold, so its channels open and a new action potential starts there.

The region just behind is still recovering (**refractory**), so the impulse can only travel one way.

Reading an oscilloscope trace

C2.2.10

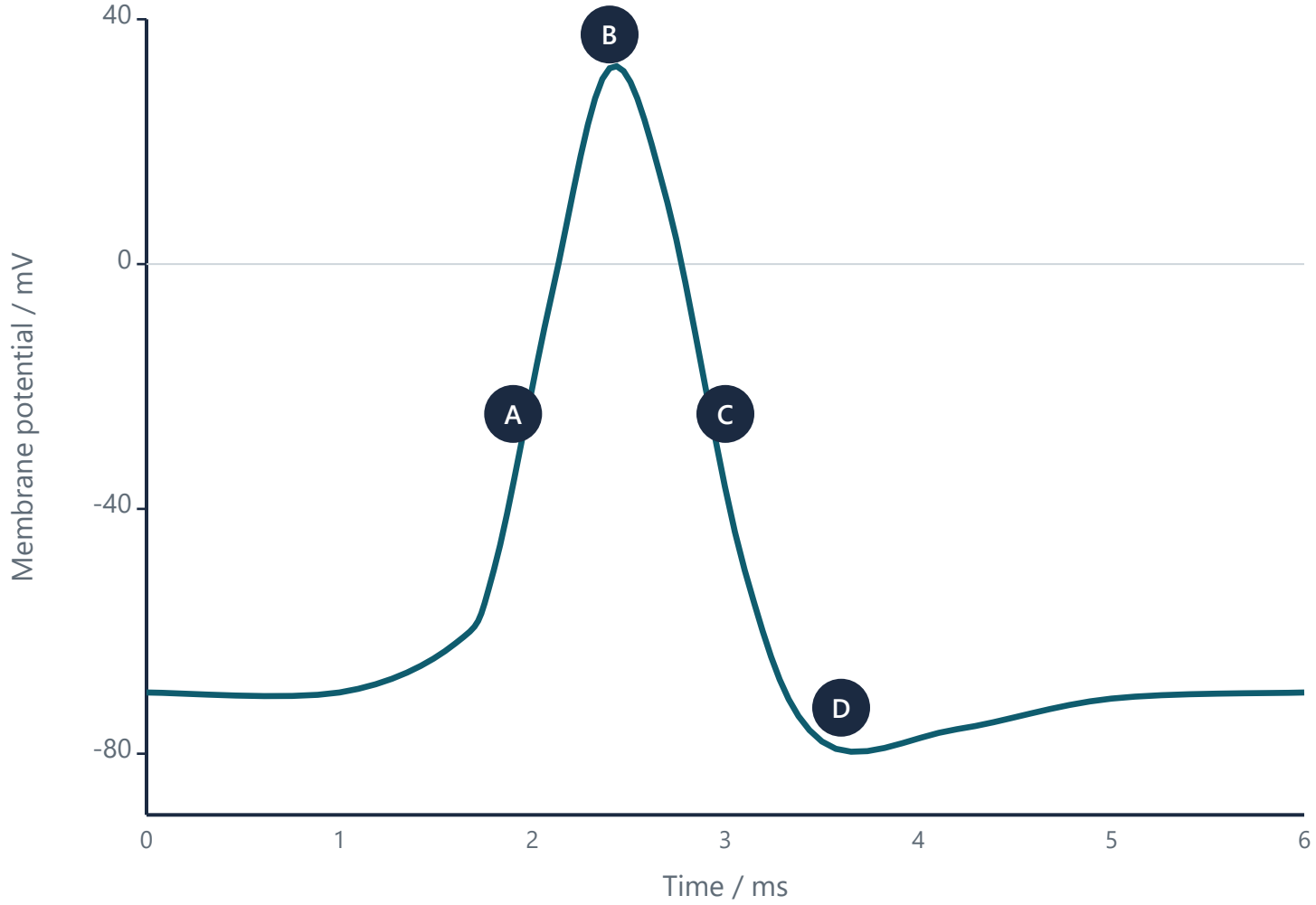


Electrodes inside and outside the axon measure the potential difference; the oscilloscope plots it against time.

Practise reading: the resting value, the peak, how long it lasts, and where each channel opens.

Practice: label the trace

C2.2.8, 10



1. For A to D, state what the Na^+ and K^+ channels are doing.
2. Read the peak voltage and the duration.
3. Explain why the trace dips below −70 mV at D.

Working with data: stimulus strength

C2.2.8, 10

Action potentials recorded from one sensory neuron with stimuli of different strength. These data are invented for practice; the pattern is realistic.

Stimulus	Peak / mV	Impulses per second
weak	none	0
moderate	+30	20
strong	+30	80

1. Explain why there is no impulse for the weak stimulus.
2. Explain why the peak is the same for moderate and strong.
3. How does the neuron signal a strong stimulus?

Check-in: class 1

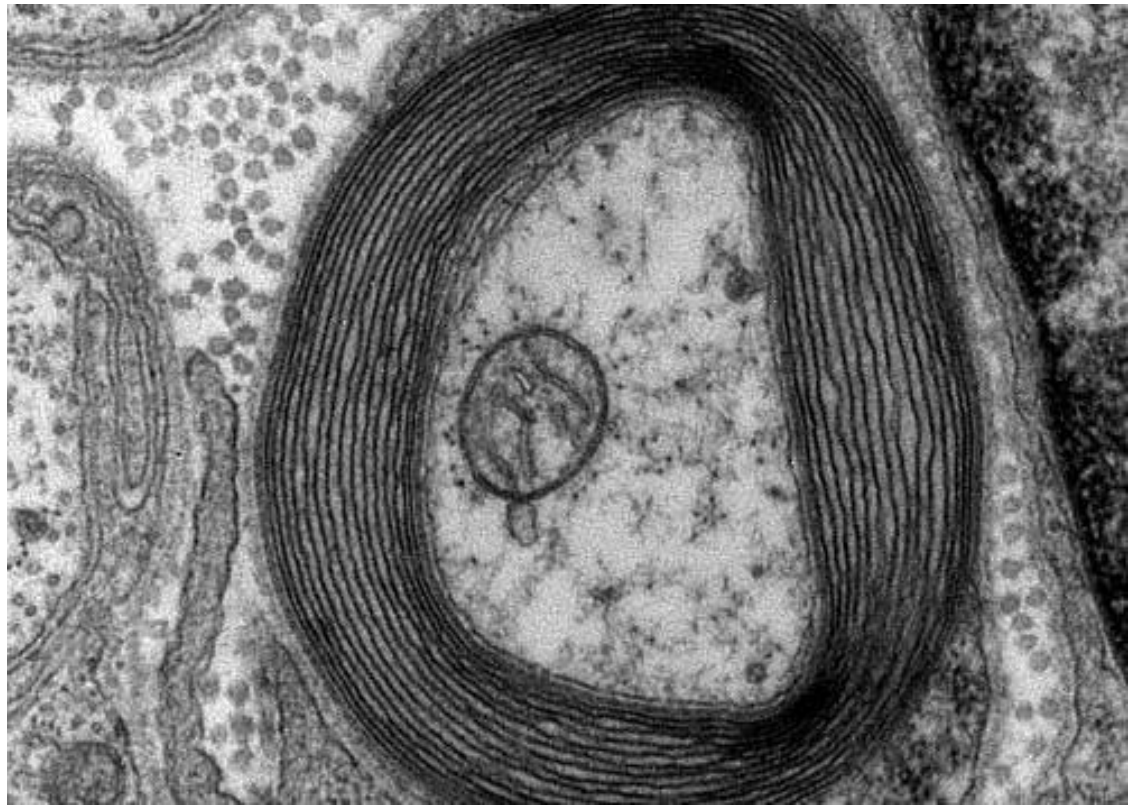
C2.2.8–10

1. State the resting potential and the threshold.
2. Explain what causes depolarization.
3. Explain what causes repolarization.
4. Explain how local currents propagate the impulse.
5. Explain why the impulse travels in one direction only.

How to use this

Answer without your notes, then check against slides 3–8.

Insulated axons



Myelinated nerve fibres (TEM). CC BY-SA 3.0.

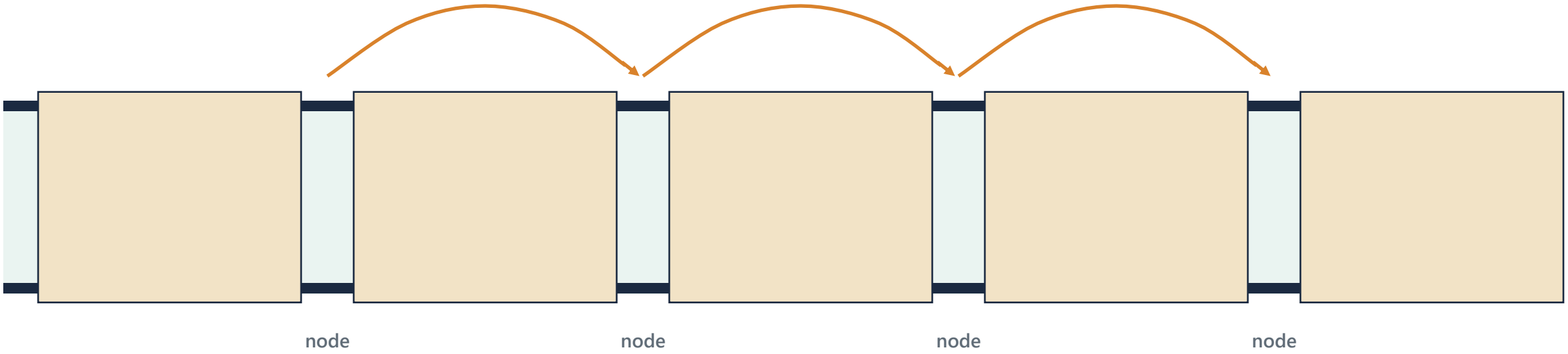
Many axons are wrapped in **myelin**: layers of membrane from Schwann cells.

Myelin insulates the axon. The gaps between Schwann cells are the **nodes of Ranvier**.

Next class: why this makes impulses much faster.

Saltatory conduction

C2.2.11



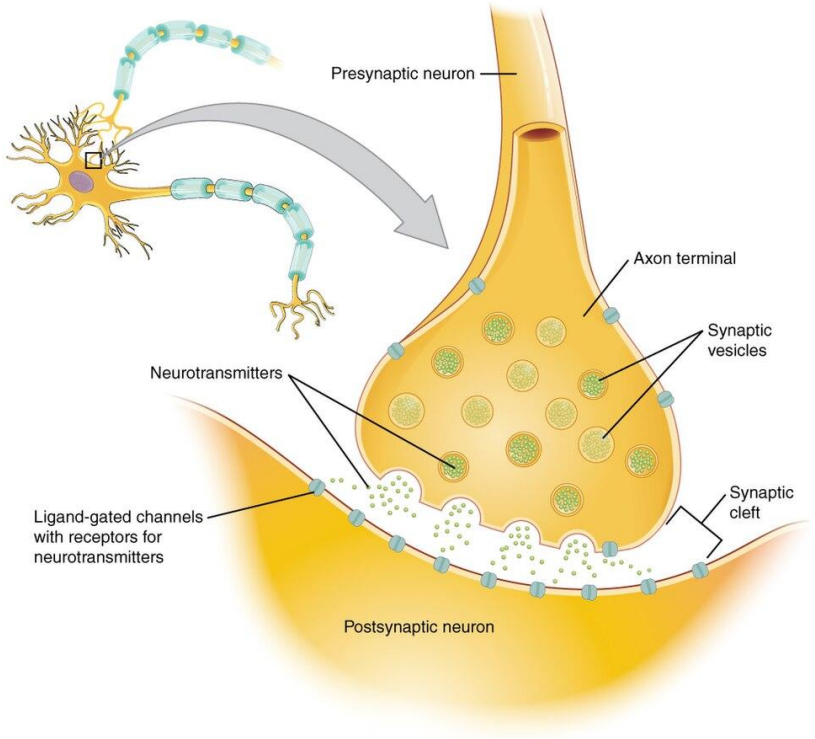
Ions cannot cross the membrane where it is covered by myelin, so action potentials only happen at the **nodes**. The impulse jumps from node to node: **saltatory conduction**.

Speed: up to about 100 m/s in myelinated fibres, compared with about 1 m/s in unmyelinated ones.

Synapses: a quick recap

C2.2.12

Impulse arrives → Ca^{2+} enters → **Vesicles release neurotransmitter** → It binds receptors → It is removed



Chemical synapse. OpenStax, CC BY 4.0.

Drugs and pesticides can change any of these steps. Two examples follow.

Neonicotinoids: blocking the receptor

C2.2.12



Honey bee. CC BY-SA 2.5.

Neonicotinoid pesticides bind to acetylcholine receptors in insect synapses. Acetylcholinesterase cannot break them down, so the receptors stay blocked.

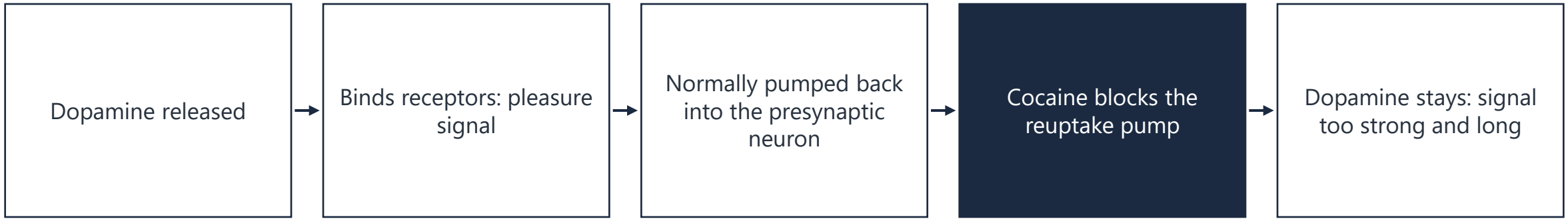
Transmission fails: the insect is paralysed and dies. They bind insect receptors much more strongly than human ones.

The problem

They also harm pollinators such as bees, so several have been banned for outdoor use in the EU.

Cocaine: blocking reuptake

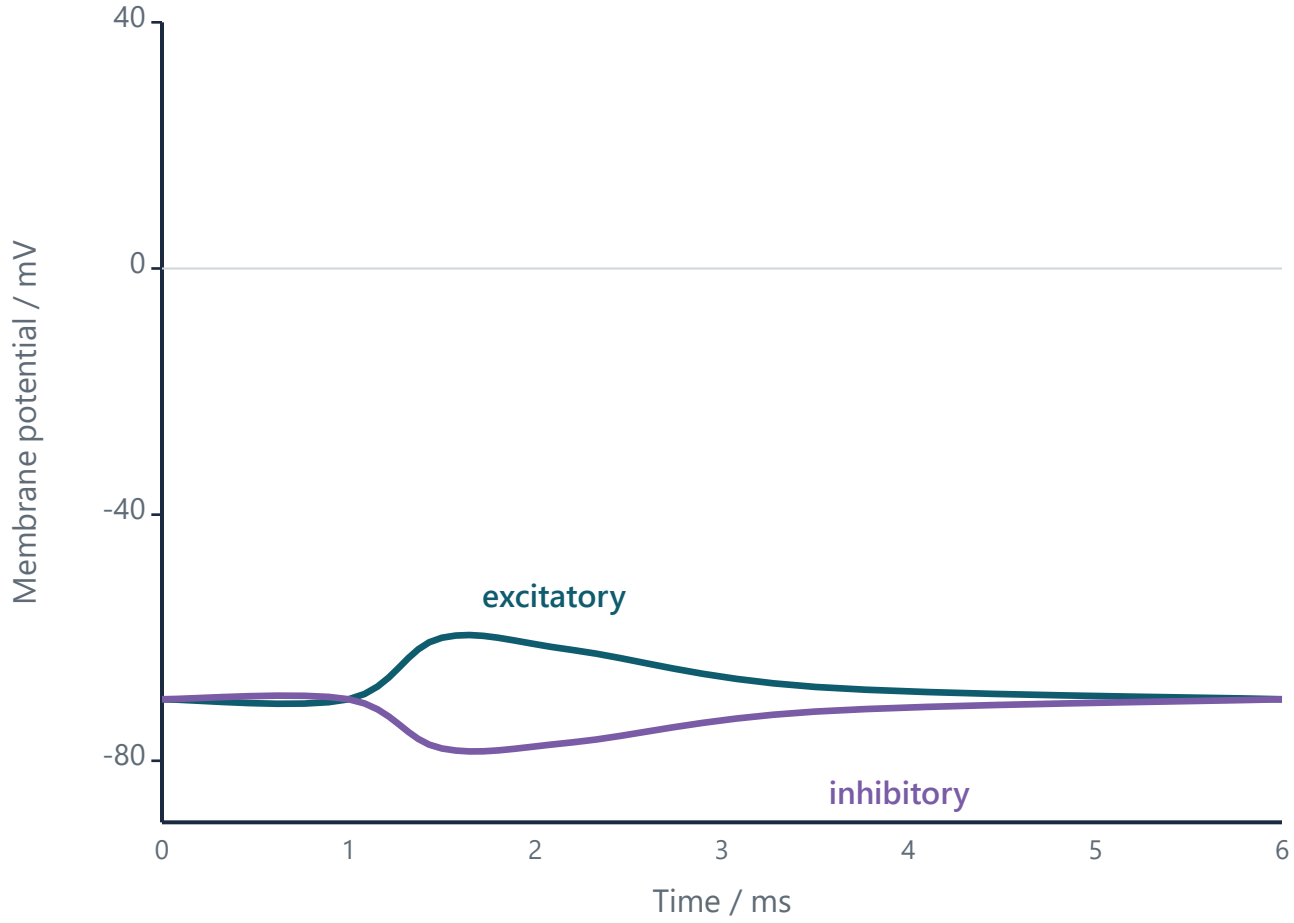
C2.2.12



Cocaine affects synapses in the brain's reward pathways that use **dopamine**. By stopping reuptake, it leaves dopamine in the synaptic cleft, giving an intense, prolonged effect.

Over time the brain makes fewer dopamine receptors, so the user needs more of the drug: tolerance and addiction.

Inhibitory synapses

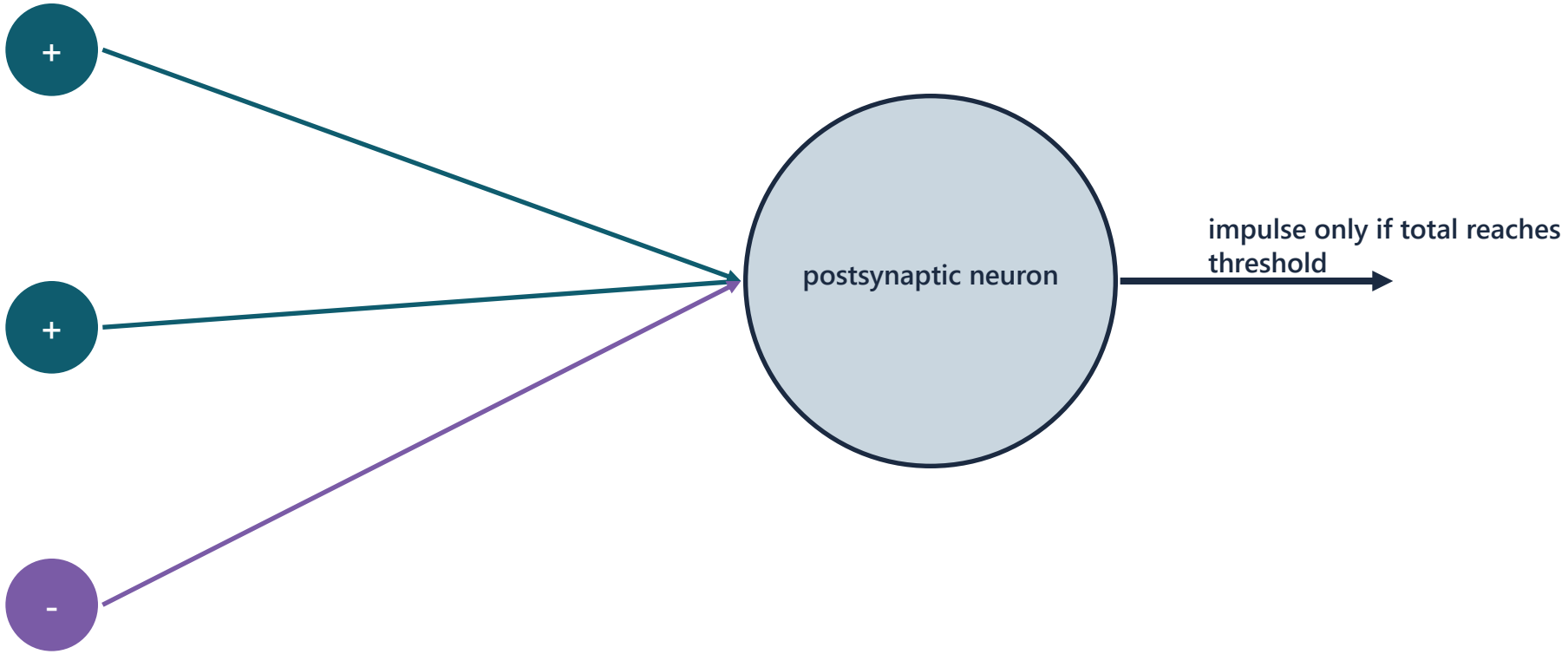


Some neurotransmitters, such as **GABA**, open channels that let chloride ions in or K^+ out.

The inside becomes **more negative**: **hyperpolarization**. This is an **inhibitory postsynaptic potential**, making an action potential less likely.

Adding up the inputs

C2.2.14



A neuron receives many synapses, some excitatory (+), some inhibitory (-). It **sums** them: if the total reaches threshold, it fires. This is how neurons make decisions.

Practice: predict the effect

C2.2.12–14

A drug blocks GABA receptors

Predict the effect on the postsynaptic neuron.

An insecticide stops acetylcholinesterase

Explain why the insect's muscles go into spasm.

Two weak excitatory inputs arrive together

Explain how they can trigger an impulse.

How to use this

Answer each in two or three sentences.

Working with data: conduction speed

C2.2.11

Conduction speed in different nerve fibres (rounded typical values).

Fibre	Diameter / μm	Speed / m s^{-1}
Unmyelinated (pain)	1	1
Squid giant axon (unmyelinated)	500	25
Myelinated (touch)	10	50
Myelinated (motor)	15	100

1. Compare the squid axon with the myelinated motor fibre.
2. Explain the difference.
3. Suggest an advantage of myelin for a large animal.

Check-in: class 2

C2.2.11–14

1. Explain how myelin speeds up conduction.
2. Explain how neonicotinoids kill insects.
3. Explain how cocaine affects dopamine synapses.
4. Describe an inhibitory postsynaptic potential.
5. Explain summation.

How to use this

Answer without your notes, then check against slides 11–18.

Detecting pain



Chilli peppers. CC BY-SA 3.0.

Pain receptors are **free nerve endings**: bare neuron tips in the skin with ion channels that open with heat, acid, damage or certain chemicals.

Capsaicin in chilli opens the same channels as heat: that is why chilli feels hot.

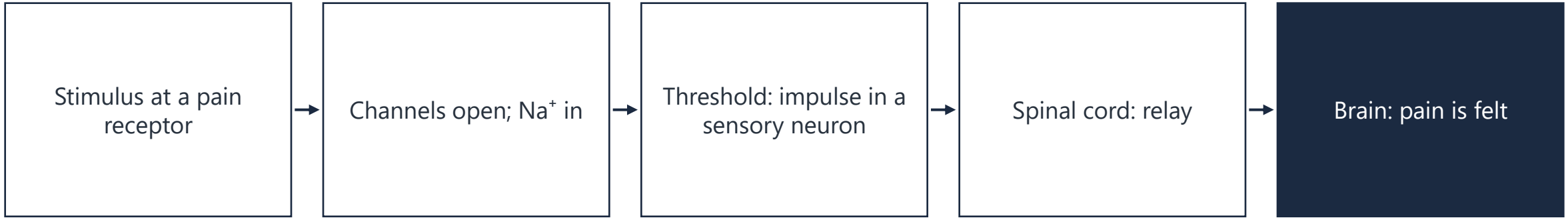
Na^+ enters, the ending depolarizes, and impulses travel to the brain, which feels pain.

Why pain matters

It warns of damage. People born unable to feel pain often injure themselves badly without noticing.

From skin to brain

C2.2.15

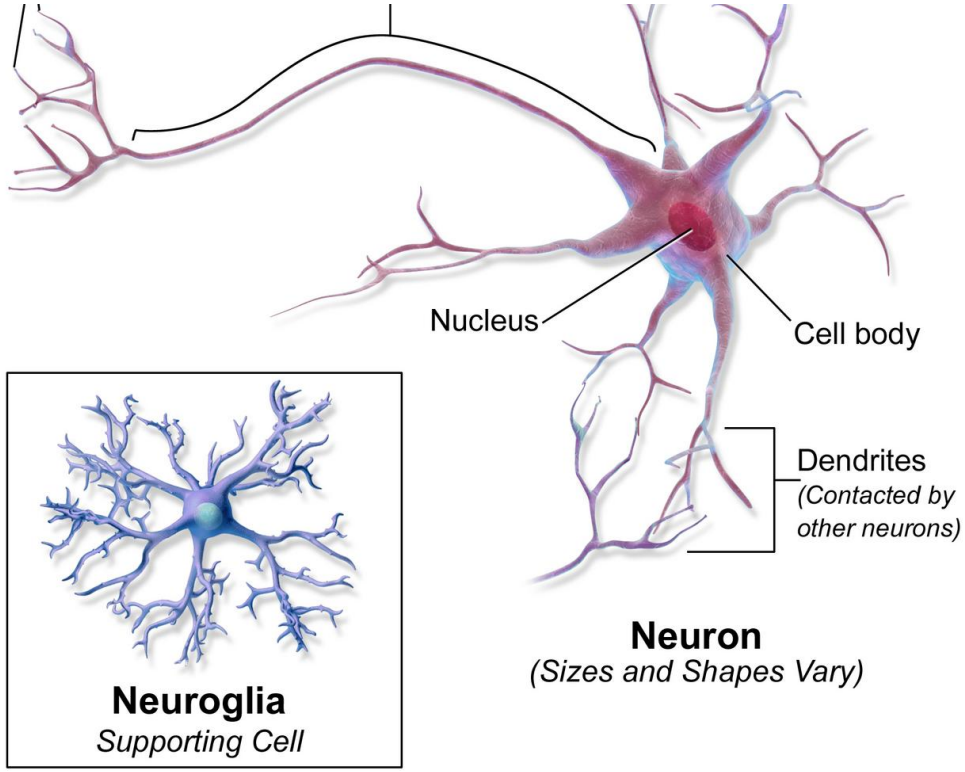


Pain is only felt when signals reach the **brain**. That is why painkillers can act on the endings, on the spinal cord or in the brain.

Pain fibres are thin and often unmyelinated, so dull pain can arrive a moment after the first sharp sensation.

Consciousness: an emergent property

C2.2.16



Neural tissue. Blausen, CC BY 3.0.

No single neuron is conscious. Consciousness **emerges** from the activity of billions of neurons interacting through trillions of synapses.

An **emergent property** is one that the whole system has but its parts do not.

Think

Can a property that emerges from neurons be fully explained by studying neurons one at a time?

Working with data: a general anaesthetic

C2.2.14–16

Postsynaptic responses with and without an anaesthetic that strengthens inhibitory synapses. These data are invented for practice.

Measurement	Without	With
IPSP size / mV	-5	-12
Impulses per second	40	6
Patient response to voice	yes	no

1. Describe the effect on IPSPs.
2. Explain the fall in impulses.
3. Suggest how this links to loss of consciousness.

Common mistakes to avoid

Not this	But this
The pump causes depolarization.	Na^+ channels opening cause depolarization.
A stronger stimulus gives a bigger action potential.	All or nothing: same size, but more frequent.
Impulses travel along myelin.	Action potentials happen only at the nodes.
Inhibitory synapses stop the neuron completely.	They make firing less likely; summation decides.

Check that you can...

1. **Explain** depolarization and repolarization
2. **Explain** how local currents propagate an action potential
3. **Analyse** oscilloscope traces
4. **Explain** saltatory conduction
5. **Explain** the effects of neonicotinoids and cocaine
6. **Explain** inhibitory postsynaptic potentials and summation
7. **Outline** how pain is detected
8. **Discuss** consciousness as an emergent property

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain what causes depolarization during an action potential.
2. Explain why myelinated fibres conduct faster.
3. Outline how cocaine affects transmission at a synapse.

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